

## 4-2c Shifts in the Demand Curve

Because the market demand curve holds other things constant, it need not be stable over time. If something happens to alter the quantity demanded at any given price, the demand curve shifts. For example, suppose the American Medical Association discovers that people who regularly eat ice cream live longer, healthier lives. The discovery would raise the demand for ice cream. At any given price, buyers would now want to purchase a larger quantity of ice cream, and the demand curve for ice cream would shift.

Figure 3 illustrates shifts in demand. Any change that increases the quantity demanded at every price, such as our imaginary discovery by the American Medical Association, shifts the demand curve to the right and is called an *increase in demand*. Any change that reduces the quantity demanded at every price shifts the demand curve to the left and is called a *decrease in demand*.

### Figure 3

#### Shifts in the Demand Curve

Any change that raises the quantity that buyers wish to purchase at any given price shifts the demand curve to the right. Any change that lowers the quantity that buyers wish to purchase at any given price shifts the demand curve to the left.

Changes in many variables can shift the demand curve. Let's consider the most important.

#### Income

What would happen to your demand for ice cream if you lost your job one summer? Most likely, your demand would fall. A lower income means that you have less to spend in total, so you would have to spend less on some—and probably most—goods. If the demand for a good falls when income falls, the good is called a

**normal good** (a good for which an increase in income raises the quantity demanded) .

Normal goods are the norm, but not all goods are normal goods. If the demand for a good rises when income falls, the good is called an **inferior good**. An example of an inferior good might be bus rides. As your income falls, you are less likely to buy a car or take a cab and more likely to ride a bus.

### Prices of Related Goods

Suppose that the price of frozen yogurt falls. The law of demand says that you will buy more frozen yogurt. At the same time, you will probably buy less ice cream. Because ice cream and frozen yogurt are both cold, sweet, creamy desserts, they satisfy similar desires. When a fall in the price of one good reduces the demand for another good, the two goods are called

**substitutes** (two goods for which an increase in the price of one leads to an increase in the demand for the other)

. Substitutes are often pairs of goods that are used in place of each other, such as hot dogs and hamburgers, sweaters and sweatshirts, and movie tickets and film streaming services.

Now suppose that the price of hot fudge falls. According to the law of demand, you will buy more hot fudge. Yet in this case, you will likely buy more ice cream as well because ice cream and hot fudge are often consumed together. When a fall in the price of one good raises the demand for another good, the two goods are called

**complements** (two goods for which an increase in the price of one leads to a decrease in the demand for the other)

. Complements are often pairs of goods that are used together, such as gasoline and automobiles, computers and software, and peanut butter and jelly.

### Tastes

Perhaps the most obvious determinant of your demand for any good or service is your tastes. If you like ice cream, you buy more of it. Economists normally do not try to explain people's tastes because tastes are based on historical and psychological forces that are beyond the realm of economics. Economists do, however, examine what happens when tastes change.

### Expectations

Your expectations about the future may affect your demand for a good or service today. If you expect to earn a higher income next month, you may choose to save less now and spend more of your current income on ice cream. If you expect the price of ice cream to fall tomorrow, you may be less willing to buy an ice-cream cone at today's price.

### Number of Buyers

In addition to the preceding factors, which influence the behavior of individual buyers, market demand depends on the number of these buyers. If Peter were to join Catherine and Nicholas as another consumer of ice cream, the quantity demanded in the market would be higher at every price, and market demand would increase.

Summary

The demand curve shows what happens to the quantity demanded of a good as its price varies, holding constant all the other variables that influence buyers. When one of these other variables changes, the quantity demanded at each price changes, and the demand curve shifts. [Table 1](#) lists the variables that influence how much of a good consumers choose to buy.

Table 1

Variables That Influence Buyers

This table lists the variables that affect how much of any good consumers choose to buy. Notice the special role that the price of the good plays: A change in the good’s price represents a movement along the demand curve, whereas a change in one of the other variables shifts the demand curve.

| Variable                 | A Change in This Variable . . .              |
|--------------------------|----------------------------------------------|
| Price of the good itself | Represents a movement along the demand curve |
| Income                   | Shifts the demand curve                      |
| Prices of related goods  | Shifts the demand curve                      |
| Tastes                   | Shifts the demand curve                      |
| Expectations             | Shifts the demand curve                      |
| Number of buyers         | Shifts the demand curve                      |

If you have trouble remembering whether you need to shift or move along the demand curve, it helps to recall a lesson from the appendix to [Chapter 2](#). A curve shifts when there is a change in a relevant variable that is not measured on either axis. Because the price is on the vertical axis, a change in price represents a movement along the demand curve. By contrast, income, the prices of related goods, tastes, expectations, and the number of

buyers are not measured on either axis, so a change in one of these variables shifts the demand curve.

### Case Study: Two Ways to Reduce Smoking

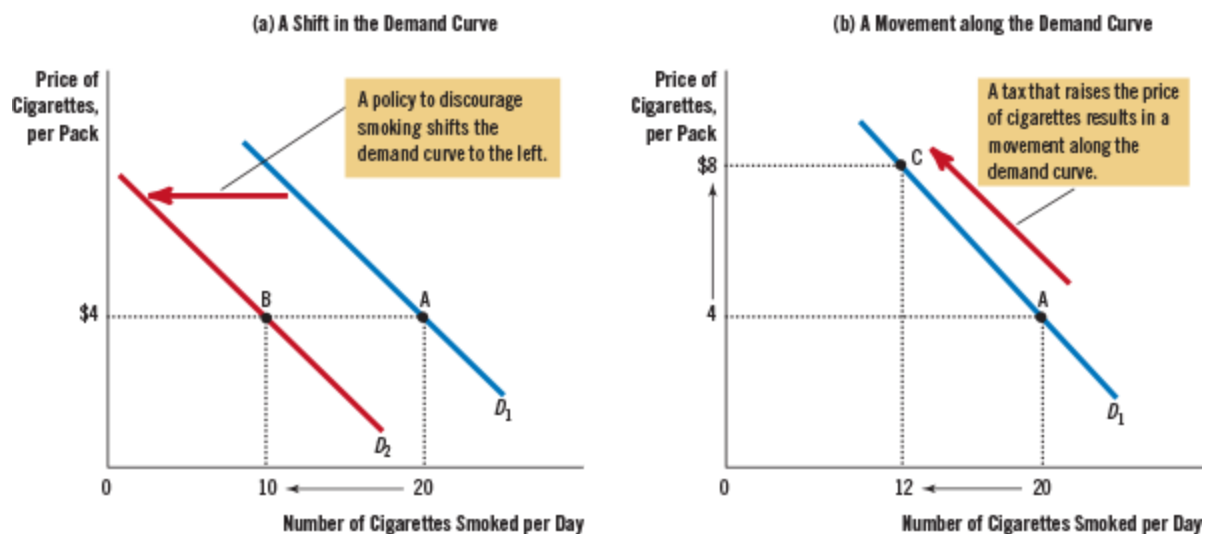
Because smoking can lead to various illnesses, policymakers often want to reduce the amount that people smoke. There are two ways that they can attempt to achieve this goal.

One way to reduce smoking is to shift the demand curve for cigarettes and other tobacco products. Public service announcements, mandatory health warnings on cigarette packages, and the prohibition of cigarette advertising on television are all policies aimed at reducing the quantity of cigarettes demanded at any given price. If successful, these policies shift the demand curve for cigarettes to the left, as in panel (a) of Figure 4.

**Figure 4**

### Shifts in the Demand Curve versus Movements along the Demand Curve

If warnings on cigarette packages convince smokers to smoke less, the demand curve for cigarettes shifts to the left. In panel (a), the demand curve shifts from  $D_1$  to  $D_2$ . At a price of \$4 per pack, the quantity demanded falls from 20 to 10 cigarettes per day, as reflected by the shift from point A to point B. By contrast, if a tax raises the price of cigarettes, the demand curve does not shift. Instead, we observe a movement to a different point on the demand curve. In panel (b), when the price rises from \$4 to \$8, the quantity demanded falls from 20 to 12 cigarettes per day, as reflected by the movement from point A to point C.



Alternatively, policymakers can try to raise the price of cigarettes. If the government taxes the manufacture of cigarettes, for example, cigarette companies pass much of this tax on to consumers in the form of higher prices. A higher price encourages smokers to reduce the number of cigarettes they smoke. In this case, the reduced amount of smoking does not represent a shift in the demand curve. Instead, it represents a movement along the same demand curve to a point with a higher price and lower quantity, as in panel (b) of Figure 4.

How much does the amount of smoking respond to changes in the price of cigarettes? Economists have attempted to answer this question by studying what happens when the tax on cigarettes changes. They have found that a 10 percent increase in the price causes a 4 percent reduction in the quantity demanded. Teenagers are especially sensitive to the price of cigarettes: A 10 percent increase in the price causes a 12 percent drop in teenage smoking.

A related question is how the price of cigarettes affects the demand for other products, such as marijuana. Opponents of cigarette taxes often argue that tobacco and marijuana are substitutes so that high cigarette prices encourage marijuana use. By contrast, many experts on substance abuse view tobacco as a “gateway drug” leading young people to experiment with other harmful substances. Most studies of the data are consistent with this latter view: They find that lower cigarette prices are associated with greater use of marijuana. In other words, tobacco and marijuana appear to be complements rather than substitutes.

Pablo del Rio Sotelo/ [Shutterstock.com](https://www.shutterstock.com).

#### QuickQuiz

4. A change in which of the following will NOT shift the demand curve for hamburgers?
  - a. the price of hot dogs
  - b. the price of hamburgers
  - c. the price of hamburger buns
  - d. the income of hamburger consumers
  
5. Which of the following will shift the demand curve for pizza to the right?
  - a. an increase in the price of hamburgers, a substitute for pizza

- b. an increase in the price of root beer, a complement to pizza
- c. the departure of college students, as they leave for summer vacation
- d. a decrease in the price of pizza

6. If pasta is an inferior good, then the demand curve shifts to the \_\_\_\_ when \_\_\_\_ rises.

- a. right; the price of pasta
- b. right; consumers' income
- c. left; the price of pasta
- d. left; consumers' income

Chapter 4: The Market Forces of Supply and Demand: 4-2c Shifts in the Demand Curve

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.